

Lab-4 VibeCoding — Chat History

Project: Word Scramble Pygame Game

Repository: 62_scramble

Purpose: Record the prompts, implementation steps, testing, and debugging carried out during the VibeCoding assignment.

Assignment Context

The Lab-4 assignment required fixing the deliberate bug in the Word Scramble game and implementing the three additional features listed in the project README: a HINT button, a countdown timer, and graphical letter tiles. The final deliverables include the before video, after video, updated code, and exported chat history.

Task 1 — Fix Guess Comparison Bug

Prompt used: Fix the broken guess comparison so that the player's guess is compared with the secret word rather than the already-scrambled word. Preserve the existing game behavior.

Change made:

```
Before: is_correct = (guess == self.scrambled_word)
After:  is_correct = (guess == self.secret_word)
```

Testing: A scrambled word such as ROCKET was tested. The corrected answer was accepted after changing the comparison to self.secret_word.

Task 2 — Add HINT Button

Prompt used: Add a clickable HINT button beside SUBMIT. When clicked, reveal the first unrevealed letter of self.secret_word in its correct position, deduct a small score penalty, never allow the score below zero, and reset revealed letters for each new round. Keep the existing submit functionality.

Implementation: Added revealed_letters, a hint penalty, a HINT button, hint-click handling, and display of revealed letters/underscores.

Testing: The HINT button was tested and the revealed-letter behavior and score penalty were retained while continuing to support the game.

Task 3 — Add 20-Second Countdown Timer

Prompt used: Add a 20-second countdown timer to each round. Display the remaining time, reset it for every round, keep correct guesses working, and when the timer reaches zero reveal the correct word, display TIME'S UP!, and move to the next round. Preserve Tasks 1 and 2.

Implementation: Added round_time, time_left, and pygame.time.get_ticks() based timing. The timer is displayed in the game and resets in next_round().

Testing: The timer was observed counting down during gameplay and continuing to work after the other features were added.

Task 4 — Graphical Letter Tiles

Prompt used: Replace the static scrambled-letter text with individual graphical square letter tiles. Allow the player to rearrange the tiles by dragging or clicking. Keep the existing SUBMIT, HINT, score, and timer functionality. Validate the word formed by the tile arrangement against self.secret_word.

Initial implementation: Individual square tiles were displayed. The first interaction approach attempted tile dragging, followed by a clickable tile-selection and swap approach because the README permits clickable letter

sorting.

Debugging: The first tile-swap implementation appeared to select tiles but the letters did not remain swapped. The cause was identified: `create_tiles()` was resetting `self.tile_letters` to the original scrambled word every time it was called. The function was corrected so it only recalculates tile rectangles, while `tile_letters` is initialized only when `next_round()` starts.

Final behavior: Click one tile to select it, then click another tile to swap their positions. The selected tile is highlighted. SUBMIT validates the current tile arrangement.

Testing: Tile selection and swapping were tested successfully. The corrected implementation retained the HINT and countdown timer features.

Final Implementation Summary

At the end of the coding process, the game contains all four requested changes: the guess-comparison bug is fixed, HINT functionality is available, each round has a 20-second countdown, and the scrambled letters are represented as individually clickable graphical tiles that can be swapped to form the answer.

Final Testing Checklist

- Task 1: Correct answer is compared with `self.secret_word`.
- Task 2: HINT reveals an unrevealed letter and applies the score penalty.
- Task 3: Timer counts down from 20 seconds and resets for a new round.
- Task 4: Letter tiles can be selected and swapped.
- SUBMIT checks the current tile arrangement.
- Incorrect arrangements display WRONG GUESS! Try again.
- The final submission requires one before video and one after video.

End of Chat History